

FINAL PROJECT PROGRESS

*INTEGRATED AI APPROACH FOR TELECOMMUNICATION CUSTOMER
CHURN ANALYSIS: A COMPARISON OF SUPERVISED CLASSIFICATION*

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WHAT WE HAVE ACCOMPLISHED SO FAR:

- Identified dataset imbalance and applied solutions.
- Full data cleaning and preprocessing pipeline.
- Feature engineering (categorical, numerical, binary split).
- Train-test split & transformer pipelines.
- SMOTE applied on supervised learning pipeline.
- Implemented 3 supervised models:
 - Decision Tree
 - Naive Bayes
 - MLP Classifier (ANN)
- Obtained evaluation metrics & visualizations.
- Feature importance analysis for all models.



SOURCE



Dataset includes:

- Customer demographics
- Billing info
- Subscription details
- Service types
- Contract categories
- Churn status

THE DATASET

- Rows: $\pm$ 7043 (train/test split already provided)
- Features: Mixed categorical + numeric
- Target: Churn (0 or 1)

DATA CLEANING PERFORMED

1. NAN HANDLING



```
1 nan_cols = df.isna().sum()
2 nan_cols[nan_cols > 0]
```

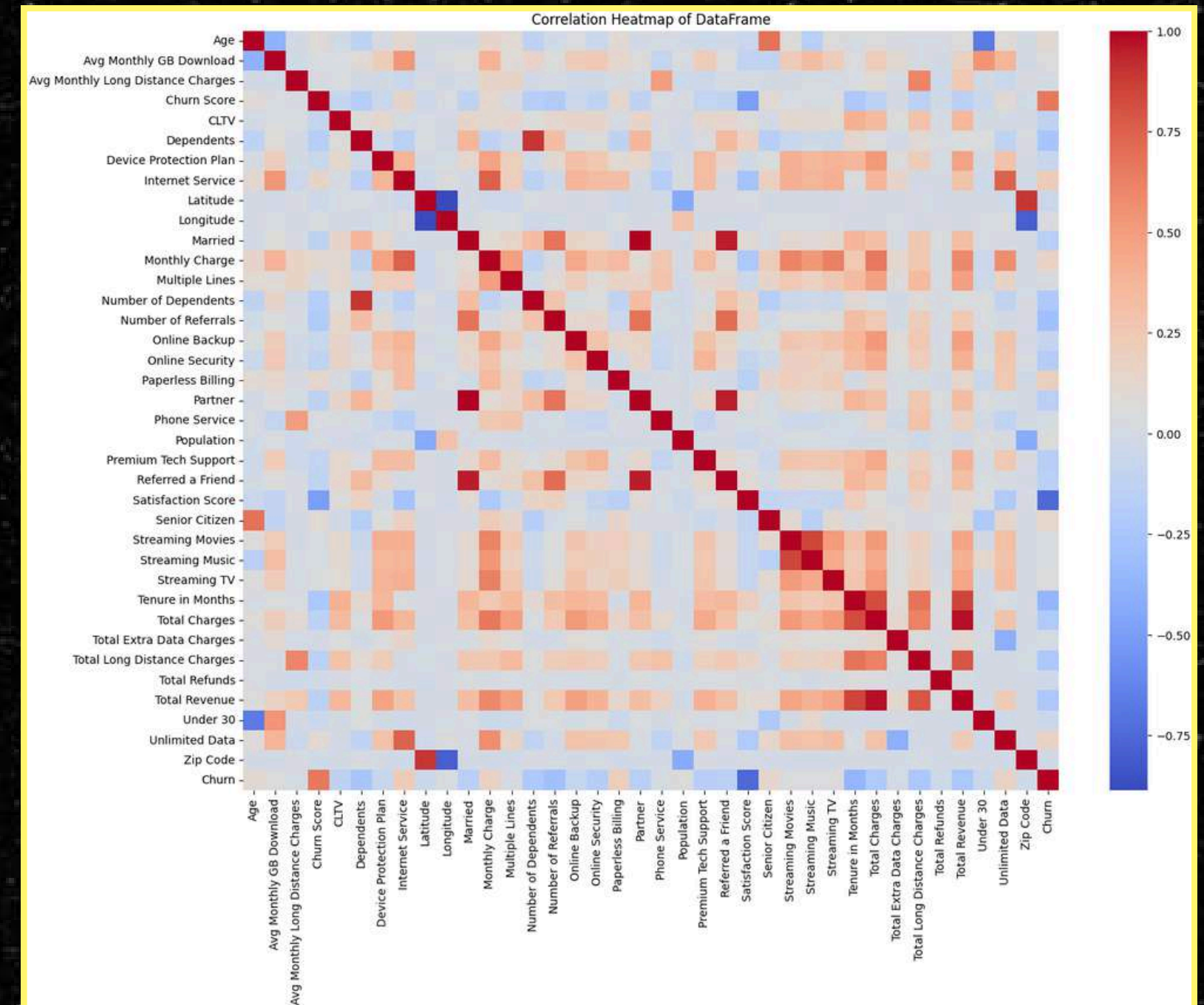
2. DROPPING IRRELEVANT / LEAKAGE FEATURES



```
1 df.drop(["Churn Category", "Churn Reason", "Customer Status",
2         "Customer ID", "Zip Code", "Churn Score",
3         "City", "Lat Long", "Latitude",
4         "Longitude", "Satisfaction Score"],
5         inplace=True, axis=1)
6
```

3. FILLING CATEGORICAL MISSING

CORRELATION HEATMAP



DATASET IMBALANCE PROBLEM

Dataset is imbalanced, especially in features where one value dominates >70%

OUTPUT:

- Several heavily imbalanced features detected.
- Columns with >90% single value were removed.

CODE USED TO DETECT IMBALANCE:

```
1 threshold = 0.7
2 biased_cols = []
3
4 for col in df.columns:
5     vc = df[col].value_counts(normalize=True)
6     max_ratio = vc.max()
7
8     if max_ratio > threshold:
9         biased_cols.append(col)
10        print(f"\n>>> Imbalanced Feature: {col}")
11        print(vc)
12        if max_ratio > 0.90:
13            df.drop(col, inplace=True, axis=1)
14            print(f"Imbalanced Feature, Dropped {col}!")
15            biased_cols.remove(col)
16
```


FEATURE ENGINEERING

1. CATEGORICAL / TEXT FEATURES

```
1 txt_cols = [col for col in df.columns if df[col].dtype == "object"]
```

2. NUMERICAL FEATURES

```
1 num_cols = [col for col in df.columns  
2             if len(df[col].unique()) > 2 and df[col].dtype != "object"]
```

3. BINARY NUMERICAL FEATURES

```
1 binary_cols = [col for col in df.columns  
2                if len(df[col].unique()) == 2 and df[col].dtype != "object"]
```

4. DISTRIBUTION VISUALIZATION FOR NUMERICAL FEATURES

Histogram per column displayed.

TRAIN-TEST SPLIT & PREPROCESSING PIPELINES

SPLIT:



```
1 X_train, X_test, y_train, y_test = train_test_split(  
2     X, y, test_size=0.2, random_state=42, stratify=y  
3 )
```

PIPELINE FOR DECISION TREE:

- OneHotEncoding only
- No scaling needed

PIPELINE FOR NAIVE BAYES & MLP:

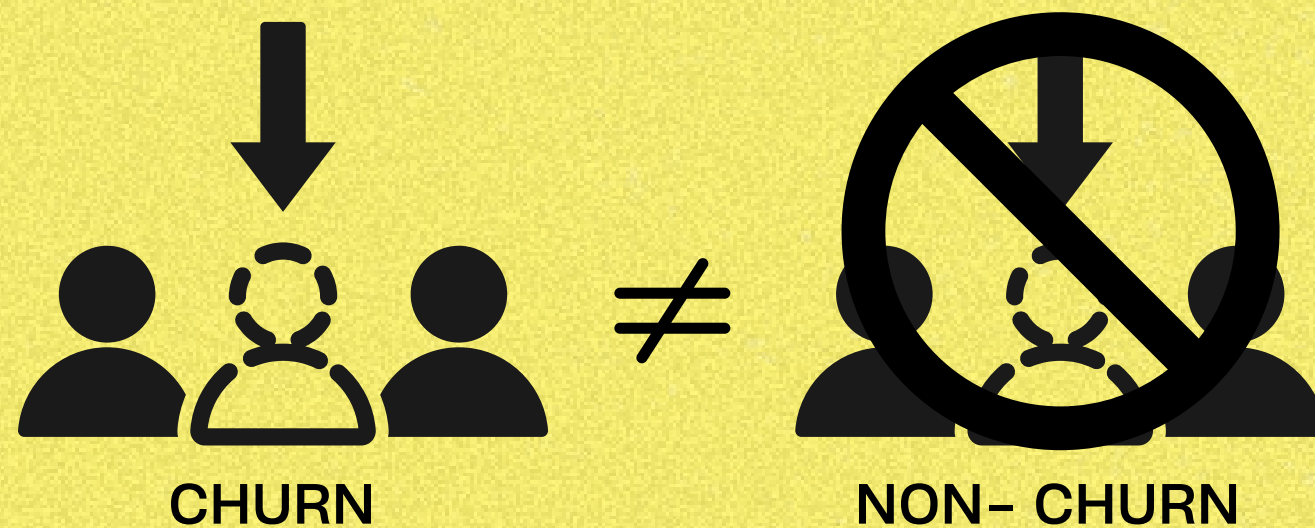
- OneHotEncoding (drop first)
- StandardScaler for numerical

Both pipelines
fitted
separately for
fairness.

DATA BALANCE HANDLING (SMOTE)

**SMOTE APPLIED ONLY ON TRAINING
DATA:**

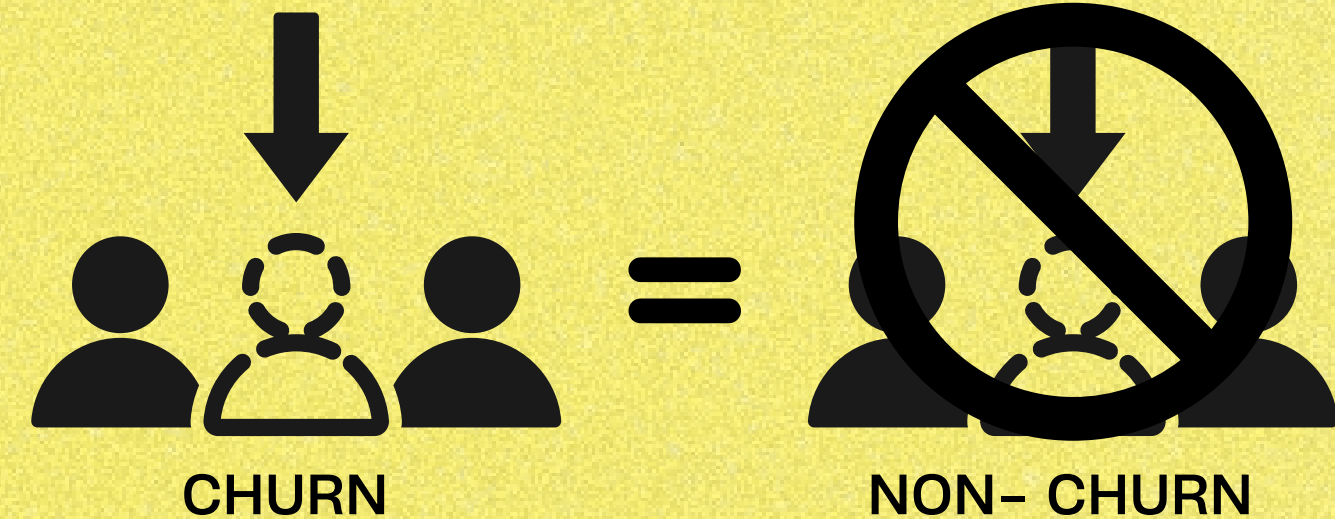
BEFORE SMOTE:



26%

74%

AFTER SMOTE:



50%

50%

DECISION TREE RESULTS

FEATURE IMPORTANCE:

- X_TRAIN: (3380, 36)
- X_TRAIN SMOTE: (4966, 31)
- X_TEST: (845, 36)

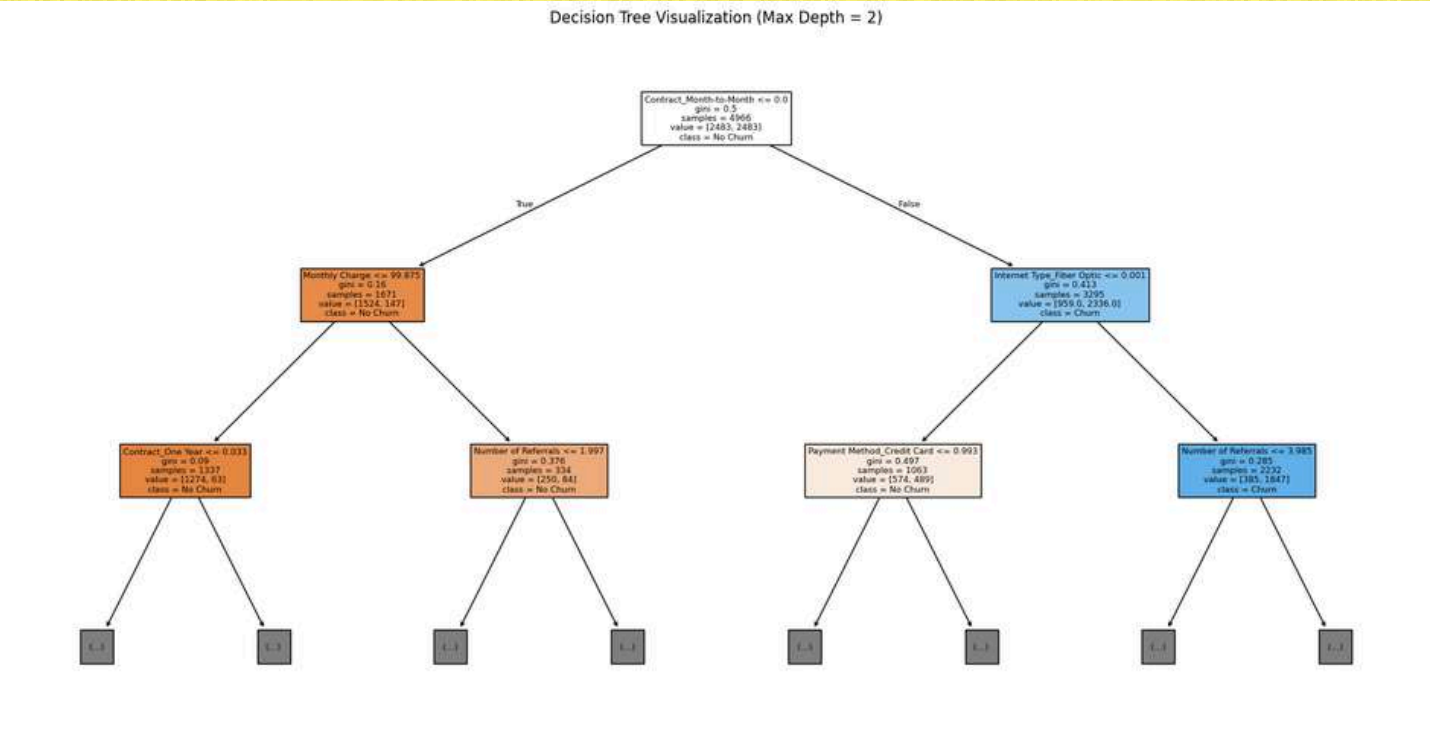
BALANCED
ACCURACY:

0.76

ACCURACY:

0.80

VISUALIZATION:

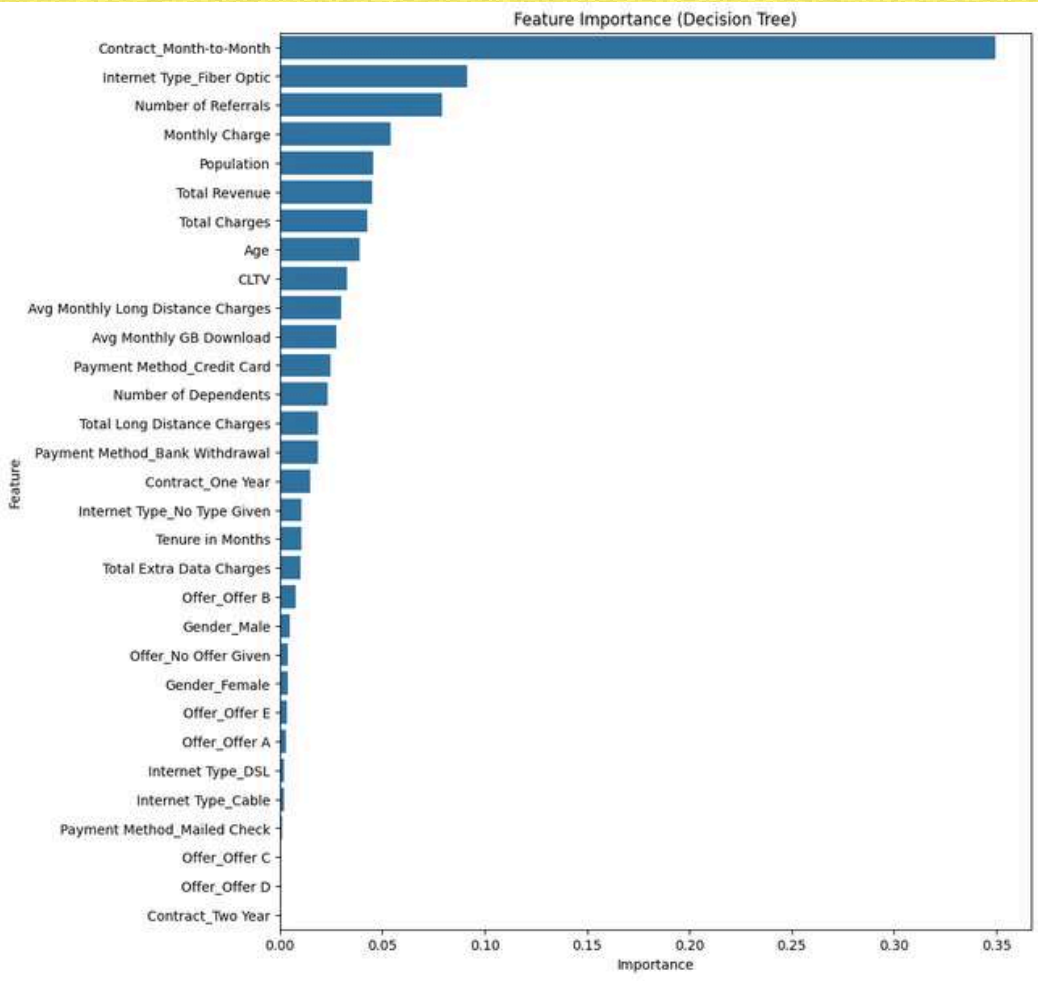


CONFUSION MATRIX:

- TRUE NON-CHURN: 523
- TRUE CHURN: 152

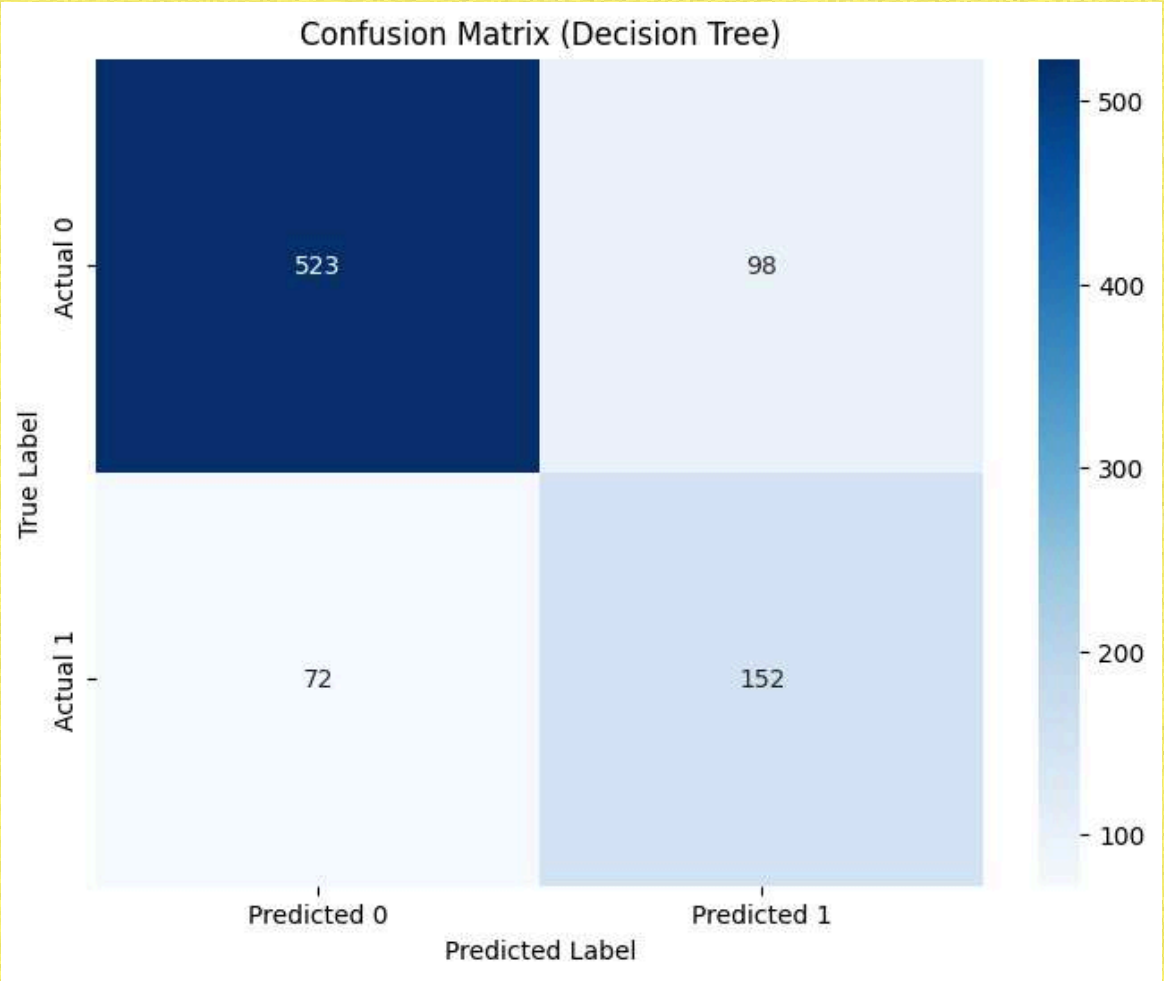
DECISION TREE VISUALIZATION

FEATURE IMPORTANCE:



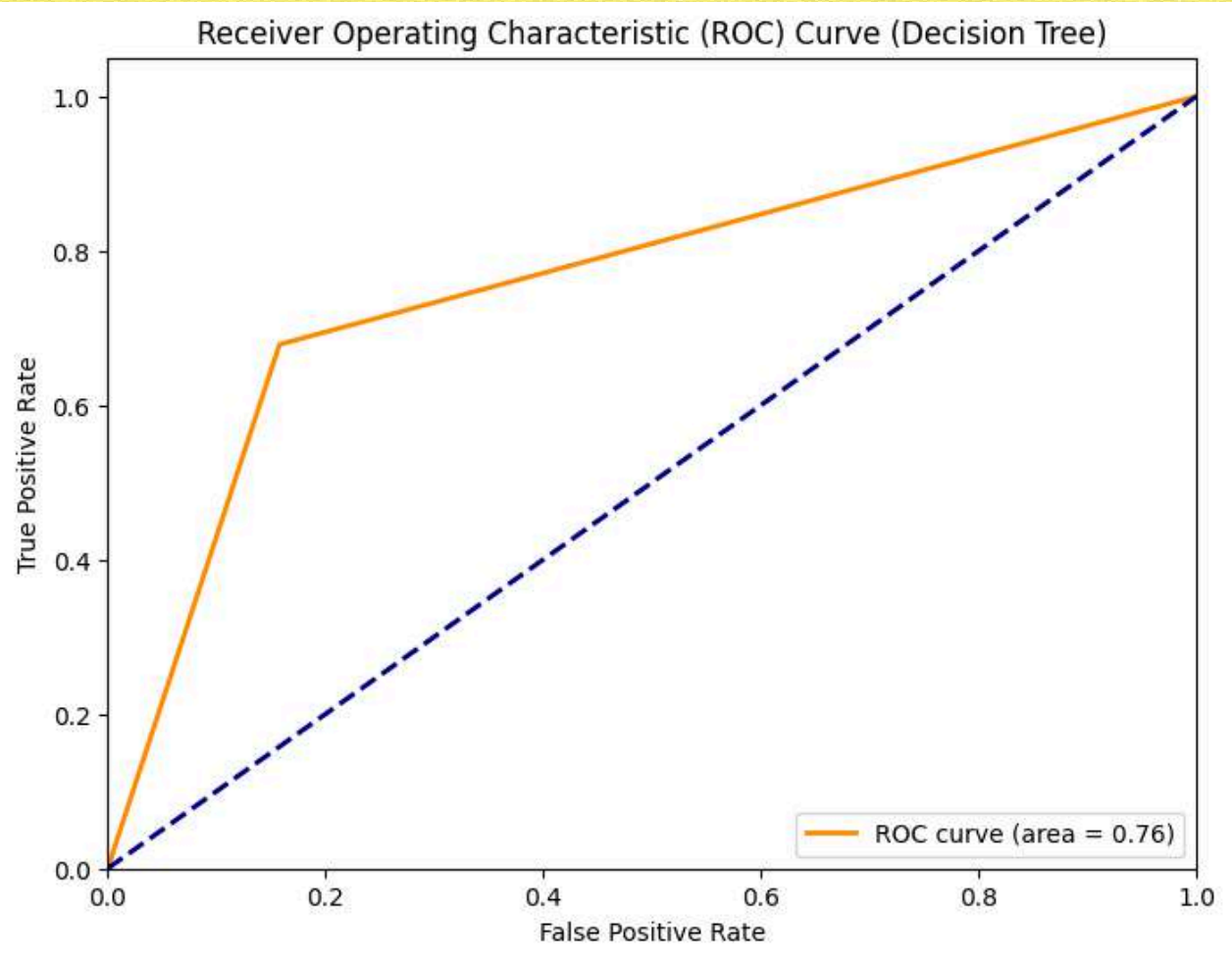
MONTH-TO-MONTH CONTRACTS EMERGED AS THE BIGGEST FACTOR CAUSING CHURN, INDICATING THAT CUSTOMERS WITHOUT LONG TERM CONTRACTS ARE MOST AT RISK OF LEAVING.

CONFUSSION MATRIX:



THE MODEL IS BETTER AT RECOGNIZING NON-CHURN THAN CHURN, BUT IT IS STILL ABLE TO CAPTURE MOST CUSTOMERS WHO ACTUALLY CHURN.

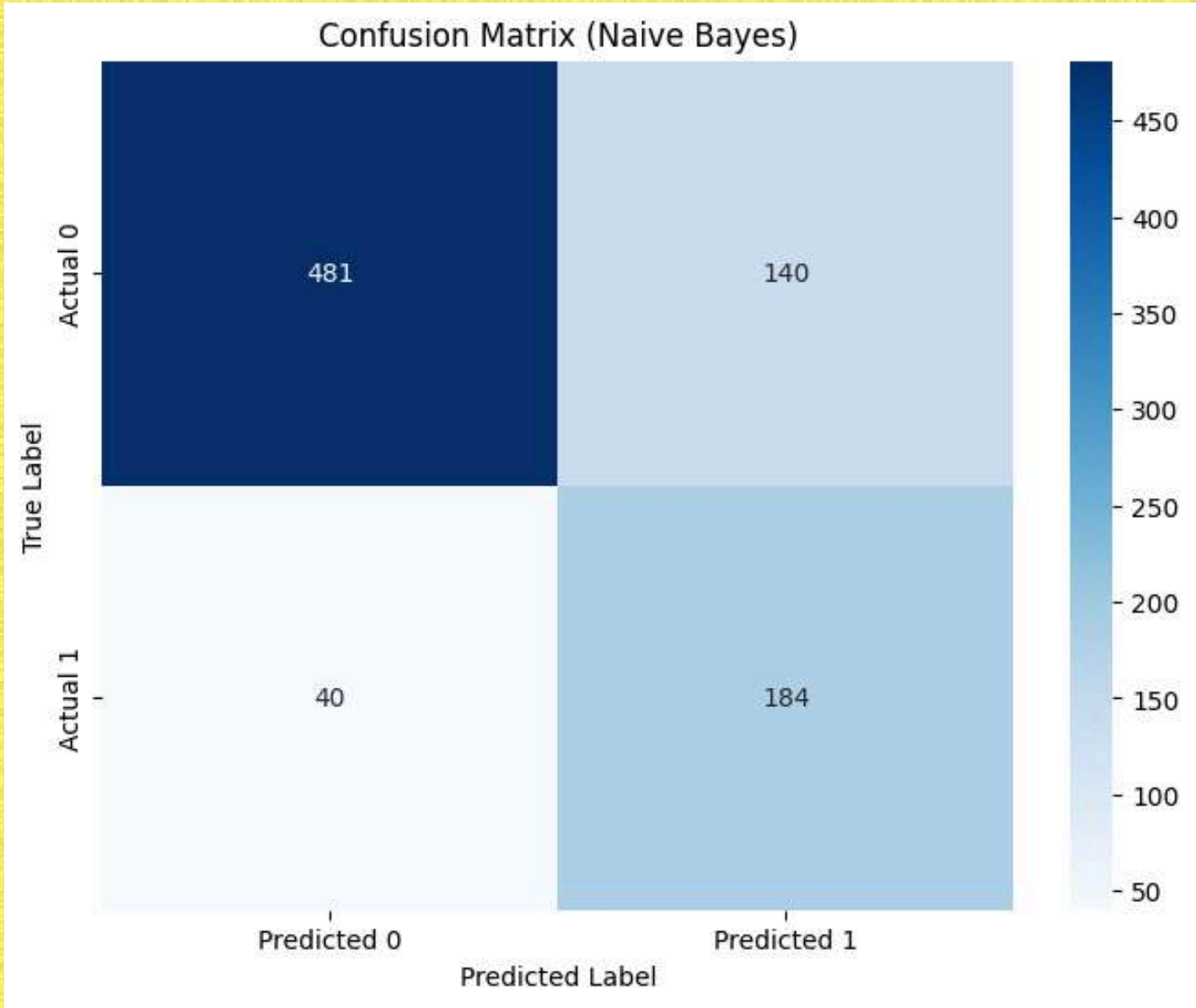
ROC CURVE DECISION TREE:



THE ROC CURVE SHOWS THAT THE MODEL HAS FAIRLY GOOD PREDICTIVE ABILITY BECAUSE IT IS ABOVE THE RANDOM LINE.

NAIVE BAYES RESULTS

CONFUSION MATRIX:



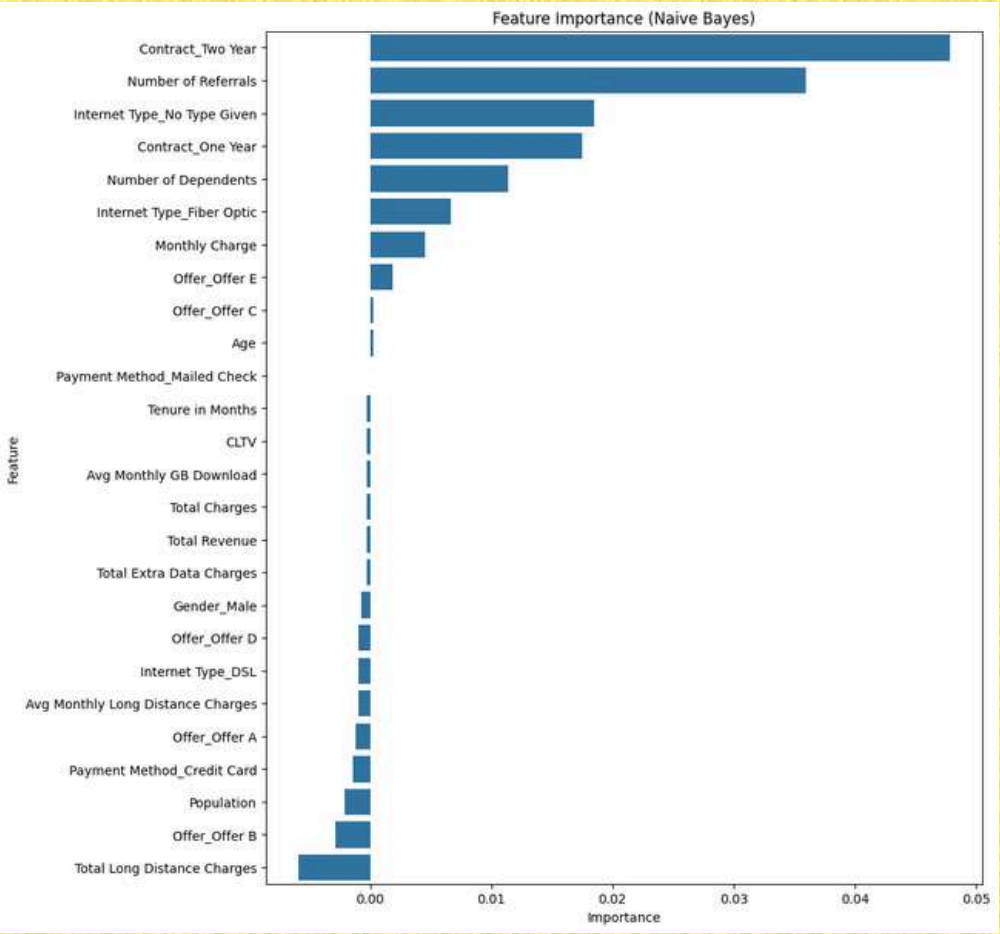
BALANCED ACCURACY:

0.7978

ACCURACY:

0.79

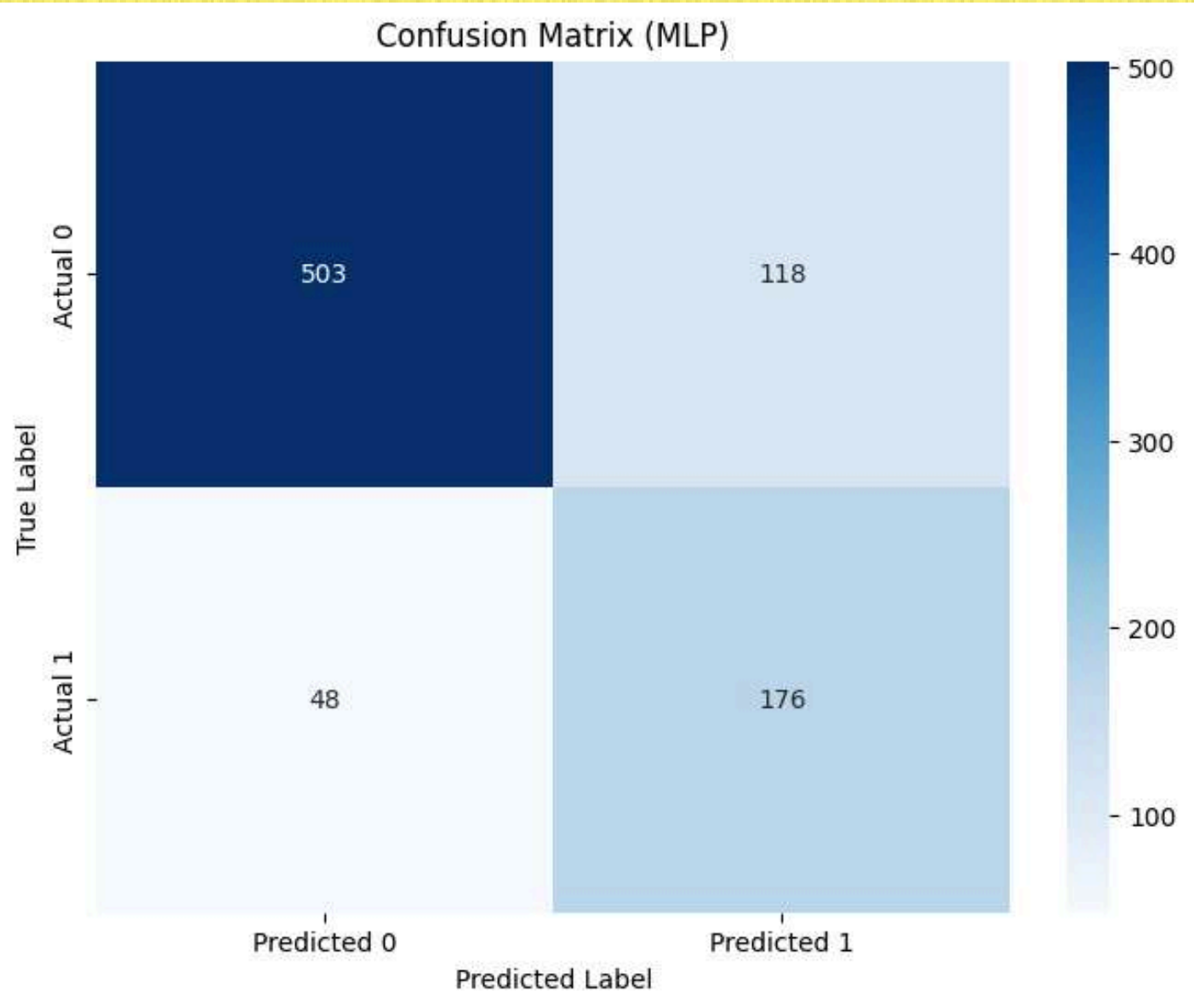
FEATURE IMPORTANCE:



**NAIVE BAYES PERFORMED BETTER AFTER SCALING AND ONE-HOT-ENCODING.*

MLP CLASSIFIER (ANN) RESULTS

CONFUSION MATRIX:



ADDS: MLP CAPTURES NONLINEAR PATTERNS THE BEST PERFORMANCE SO FAR.

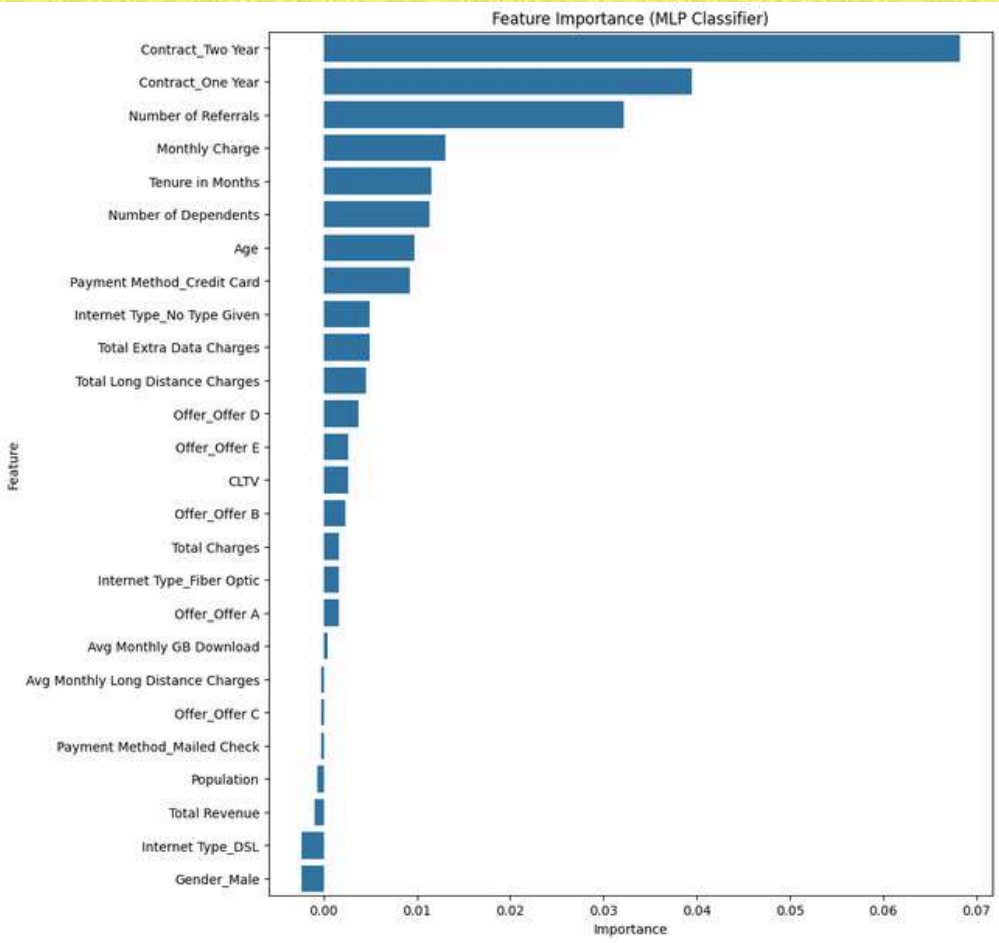
BALANCED ACCURACY:

0.7979

ACCURACY:

0.80

FEATURE IMPORTANCE:



COMPARISON IMPORTANT FEATURES

DECISION TREE:

- CONTRACT_MONTH-TO-MONTH
- INTERNET_TYPE_FIBER OPTIC
- NUMBER OF REFERRALS
- MONTHLY CHARGE

NAIVE BAYES:

CONTRACT_TWO-YEAR (HIGHEST WEIGHT)

MLP CLASSIFIER:

CONTRACT_TWO-YEAR
(STRONGEST IMPORTANCE)

CONTRACT TYPE IS THE STRONGEST CHURN DETERMINANT ACROSS MODELS.

ACCURACY COMPARISON

Model	Balance Accuracy
Decision Tree	0.760
Naive Bayes	0.7978
MLP Classifier	0.7979

MLP SLIGHTLY OUTPERFORMS THE OTHERS.

CHALLENGES & SOLUTIONS



CHALLENGES:

- DATASET IMBALANCE
- MANY IRRELEVANT/LEAKAGE COLUMNS
- HIGH-CARDINALITY CATEGORICAL FEATURES
- SCALING REQUIRED FOR SOME MODELS BUT NOT OTHERS
- SEPARATE PREPROCESSING PIPELINES NEEDED



SOLUTION:

- DROPPED >90% DOMINATED FEATURES
- APPLIED SMOTE
- CLEANED & STANDARDIZED CATEGORICAL FIELDS
- BUILT TWO PIPELINE STRUCTURES
- VALIDATED USING BALANCED ACCURACY & CONFUSION MATRICES

NEXT STEPS / PLANNING

WHAT'S NEXT:

1. FINALIZE MODEL COMPARISON

- CONSOLIDATE PERFORMANCE METRICS (BALANCED ACCURACY, PRECISION, RECALL, F1-SCORE).
- CREATE FINAL COMPARISON TABLE BETWEEN DECISION TREE, NAIVE BAYES, AND MLP.
- VALIDATE RESULT CONSISTENCY ACROSS DIFFERENT RANDOM STATES.

2. IMPROVE MODEL INTERPRETABILITY

- DEEPEN INTERPRETATION OF DECISION TREE RULES.
- ANALYZE WHY CERTAIN FEATURES DOMINATE (E.G., "CONTRACT TYPE,,").
- LINK MODEL OUTPUTS TO BUSINESS-RELATED CHURN INSIGHTS.

3. STRENGTHEN DATA PREPROCESSING SECTION

- ADD EXPLANATION FOR IMBALANCED HANDLING (SMOTE).
- ADD JUSTIFICATION FOR FEATURE REMOVAL AND ENCODING CHOICES.
- REFINE STANDARDSCALER DESCRIPTION FOR ANN AND NB.

THANK YOU.

